

Connor Savugot

B.S. Computer Engineering | North Carolina State University | May 2026

consav04@gmail.com | +1-828-541-9487 | Murphy, NC | linkedin.com/in/connorsavugot | github.com/Clem085

TECHNICAL PROFILE

Embedded firmware engineer spanning programmable power conversion, BMS integration, Embedded Linux/BSPs, board bring-up, communication protocols, and hardware-firmware validation from component selection through custom-PCB testing.

TECHNICAL SKILLS

Firmware & Software: C, C++, Python, Bash; FreeRTOS, Embedded Linux, Yocto/Arago, systemd

Platforms & Hardware: MSP430FR2355, STM32, dsPIC, TI TMS320, TI AM62; BMS, bidirectional buck-boost converters, custom PCBs, board bring-up

Interfaces & Protocols: I2C, CAN/CAN FD, SPI, UART/RS232, IPMI/IPMB, ADC, PWM, GPIO, JTAG

Test & Tooling: Git/GitLab, Code Composer Studio, KiCad, SocketCAN/PCAN, oscilloscopes, logic/protocol analyzers, through-hole/SMT soldering

Additional Languages: MATLAB, VBA, Assembly, Verilog

PROFESSIONAL EXPERIENCE

AEGIS Power Systems | Embedded Firmware Engineer

Current

- Developed firmware for custom programmable bidirectional buck-boost power systems, coordinating 24 independently configurable PWM channels, ADC feedback, and transformer-coupled input/output stages through a centralized control interface.
- Integrated BMS and peripheral ICs from datasheet, schematic, and reference-design trade studies through driver development, breakout-board prototyping, custom-PCB bring-up, and validation.
- Implemented and debugged CAN/CAN FD, I2C, SPI, and UART interfaces; isolated root causes with register-level tests, oscilloscopes, logic/protocol analyzers, known-good nodes, and controlled hardware substitutions.
- Supported TI AM62-class Linux BSPs using Yocto/Arago, device trees, systemd watchdogs, and Python/Bash diagnostics; validated IPMI/IPMB sensor and FRU access for embedded platform management.

AEGIS Power Systems | Intern

May–Aug 2024 & May–Aug 2025

- Built a restricted SSH/chroot CLI and Python/Bash tools for remotely controlling and inspecting TI Yocto/Arago embedded power hardware.
- Developed checksum-validated TMS320 flashing tools, reverse-engineered RS232 transactions, and supported FreeRTOS, serial-interface, schematic, soldering, and burn-in test debugging.

SENIOR DESIGN

EV Active Sensor Adapter | MSP430FR2355

NC State University

- Defined architecture and hardware/firmware requirements for an MSP430 EV sensor adapter with BMS and power-management hardware, sensing, custom PCBs, communications, and telemetry.
- Evaluated BMS and support ICs against electrical, protection, communications, packaging, sourcing, and integration constraints; documented tradeoffs, risks, selection rationale, and validation requirements.
- Prototyped interrupt-driven SPI and staged ADC, UART, timer, GPIO, and LCD interface validation while coordinating subsystem dependencies through test plans, design reviews, and progress documentation.

TECHNICAL LEADERSHIP & EARLIER EXPERIENCE

Embedded Systems Teaching Assistant | NC State University

- Delivered lectures and hands-on support for C/MSP430 firmware, ADC, timers, interrupts, GPIO, LCDs, and custom-PCB bring-up; diagnosed firmware, soldering, wiring, and toolchain failures while teaching systematic root-cause analysis.

Lead Instructor | Project-Based Electronics Workshop

Two Semesters

- Co-led students from project scoping through final demonstration and taught through-hole/SMT soldering theory, safety, debugging, and assembly using purpose-built instructional PCBs.

TEAM Industries | Engineering Intern

Jun–Aug 2021 & Jun–Aug 2022

- Built Excel/VBA quality tools to flag out-of-spec parts and organized 17,000+ tooling and part records linking vendors, operations, locations, and revision data for manufacturing and engineering teams.